

Fahim Ahmed Irfan

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Education

Kennesaw State University	Aug 2026 - Present
Ph.D. in Computer Science	
Missouri State University	July 2025
M.Sc. in Computer Science	
Missouri State University	July 2025
Graduate Certificate in Software Engineering	
Bangladesh University of Engineering and Technology	June 2021
B.Sc. in Naval Architecture and Marine Engineering	

Publications

Journal Paper

- **F. A. Irfan**, and R. Iqbal “An Automated Policy Generation Framework Leveraging Sensor-Actuator Mapping for IoT-Enabled Smart Spaces”, in IEEE Internet of Things Journal (IoT-J), 2025 | **Under Review**
- **F. A. Irfan**, M. A. Tanvir, M. A. R Fahad, and R. Iqbal “A Metamorphic Testing Framework for Verification and Validation of Unsupervised Sensor Grouping in Smart Spaces via Spectral Clustering”, in Journal of Software: Testing, Verification and Reliability (STVR), 2025 | Accepted | **In Press**

Conference Papers

- M. A. Tanvir, **F. A Irfan**, and R. Iqbal, “GREAF: Graph-based Real-time Anomaly Detection via Forecasting for IoT-Enabled Indoor Spaces”, in Human-Centered Sensing, Modeling, and Intelligent Systems, 2026 | **Accepted**
- **F. A. Irfan**, R. Iqbal, and S. Olson, “Actuator Transition Dependency Aware Deep Learning Approach for Actuator State Prediction,” in International Conference on Computer Communications and Networks (ICCCN), 2026 | [Paper](#)
- **F. A. Irfan**, R. Iqbal, D. Eckstein, and M. Strickland “CoRe: A Comparative Study of Transformer Models for Context Recognition in Smart Classrooms”, in IEEE Computers, Software, and Applications Conference, 2025 | [Paper](#)
- M. A. Tanvir, **F. A Irfan**, and R. Iqbal, “SEAD: Sensor Event-Based Anomaly Detection for Smart Home Automation”, in IEEE Computers, Software, and Applications Conference, 2025 | [Paper](#)
- **F. A. Irfan**, C. Behl, and R. Iqbal, “Verbal Abuse Detection from Short Conversations”, in IEEE Consumer Communications & Networking Conference, 2025 | [Paper](#)
- **F. A. Irfan**, R. Iqbal, and A. Siddiqua, “SeReIn-M: Sensor Relationship Inference in Multi-Resident Smart Homes”, in IEEE Consumer Communications & Networking Conference, 2024 | [Paper](#)
- A. Siddiqua, S. Liu, R. Iqbal, **F. A. Irfan**, L. Ross, and B. Zweerink, “TIM-MARL: Information Sharing for Multi-Agent Reinforcement Learning in Smart Environments”, in IEEE Consumer Communications & Networking Conference, 2024 | [Paper](#)
- **F. A. Irfan**, and M. Karim, “Detection of Underwater Ships Categorically Using Convolutional Neural Network and Power Efficient Remotely Operated Vehicle”, International Conference on Marine Technology, MARTEC 2022 | [Paper](#)

Poster Presentations

- “An Unsupervised Learning Approach for Sensor-Actuator Relations in Smart Homes”, 32nd Annual Einhellig Graduate Interdisciplinary Forum, 2025
- “Sensor Actuator Relationship Inference in Smart Homes”, 31st Annual Einhellig Graduate Interdisciplinary Forum, 2024

Thesis

- *Towards Automated Sensor Actuator Mapping for Dynamic Smart Spaces* | [Link](#)
 - An unsupervised approach for mapping sensors to actuators in smart spaces by analyzing time series data of occupant-triggered events. It clusters sensor-actuator pairs based on temporal patterns, eliminating manual rule-setting and proving effective across dynamic real-world deployments.

Technical Skills

Languages: Python, JavaScript, Dart, Java, C, C++, SQL, HTML, CSS, Fortran

Technologies: Django, Flask, React, Flutter, Docker, Git, Apache JMeter, Jira, LaTeX

Expertise: AI, Full Stack Web Development, Microcontrollers, Image Processing, PCB Design, UI Design (Figma, Framer)

Work Experience

Research Assistant

(Jan 2024 - Jul 2026)

Multimedia Systems and Communication (MuSyC) Lab, Computer Science, Missouri State University

- Contributed to multiple grant proposal efforts, including Education in Research (EIR), NSF CPS (Small) program submissions, by assisting with writing and identifying new research directions, while also supporting lab colleagues in their research activities.
- Voluntarily continuing ongoing research projects and grant proposal writing alongside supporting undergraduate and graduate students in **Smart Environments**.

Graduate Assistant

(Jan 2025 – Jul 2025)

Computer Science Opportunities, Development and Education in Rural Schools (CODERS)

Funded by the **U.S. Department of Education's Education Innovation and Research (EIR) Program**, the CODERS Grant (2021–2025) was awarded to **Missouri State University** to enhance STEM learning in rural schools across southern Missouri through advanced technology integration, high-quality curriculum development, and educator–student engagement initiatives.

- Conducted hands-on training sessions and assisted over 30 STEM teachers across Midwest Missouri in developing block coding, problem-solving, and computational thinking skills using **Micro:bit, Cutebot, Scratch, Smart Home Kits, Robot Dogs, and Drones**.
- Organized and judged the **CODERS Olympiad** (2024 & 2025), supporting 15+ teams and more than 60 student participants.
- Collaborated with teachers to design and refine technology-driven instructional strategies, promoting effective STEM education for K–12 learners.

Graduate Mentor

(Jan 2024 – Jul 2025)

Computer Science Research Opportunity for Smart Environments (ROSE)

Funded by the **National Science Foundation (NSF)**, the ROSE Grant (2023–2025) was awarded to **Missouri State University (MSU)** to provide a unique summer research experience and academic-year mentorship for participating K–12 STEM teachers. The program enabled teachers to work alongside MSU Computer Science faculty and graduate students to explore research challenges and develop practical solutions for next-generation smart environments.

- Mentored over 20 middle and high school STEM teachers on foundational to advanced computer science topics, including **Computational thinking, Block programming, Python programming, and machine learning**. Assisted in designing customized lesson plans for integrating CS concepts into classroom teaching.
- Guided teachers through all phases of **Research Methods**: topic selection, literature review, data collection, implementation, evaluation, and academic writing.
- Developed and maintained the official **ROSE website**. Coordinated the selection of 10 highly qualified teachers from over 400 applicants across southern Missouri. Organized and managed symposiums for teachers to present research posters, overseeing all logistical and operational arrangements.

Software Engineer

(Jan 2020 - Dec 2023)

NextGen DigiTech Ltd, Bangladesh

- Developed **NextGen MyHealth**, a cross-platform telemedicine platform (Flutter, React, Django) serving 200+ doctors and 400+ patients. Integrated authentication, microservices, and real-time notifications (Celery + Redis).
- Implemented one-to-one doctor–patient video calls using **WebRTC**.
- Supervised junior developers on **QA Tek LMS**, an online learning system (500+ users) with authentication, and attendance.
- Built an **Android HRM app** (Java, Android Studio) and a **NextGen Tower** desktop tool for wind-turbine design.

Honors and Awards

- First recipient of the *MuSyC Lab Founder's Award for Excellence*, 2025
- Awarded *Outstanding Poster Presentation* at the EIDF, 2025
- Nominated *Outstanding Graduate Student* at Missouri State University, 2025
- 16th place worldwide at European Rover Challenge, 2019

Academic Activities

- Guest Speaker in CSC703 - Research Methods at MSU
- IRB Certification
- Electrical Team Lead, BUET MARS Rover Team, Team Interplanetary 2019
- Competitive Programming - Codeforces, Leetcode, CodeSignal

Projects

- Created an automated oil flow readings for ship oil tanker using TUF-2000 flow meter.
- Ship Category Detection using CNN and Remotely Operated Vehicle (ROV) - Achieved 96% accuracy
- Industrial Automation Robot - Image processing-based tracking and sorting
- Line Follower Robot - Autonomous navigation using IR sensors